

# SEC P vs NP log 2026-05-11

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-11 11:00 UTC

## SEC P vs NP — daily log 2026-05-11

26 cycles recorded

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### Homotopy Group Rank Inverse Proportional to Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Topology (Homotopy Theory)  $\times$  Disjointness Communication Complexity
- **Recorded:** 2026-05-11 00:01 UTC
- **Entry ID:** c0f03be34ae3

#### Statement

For any bipartite graph  $G = (U \cup V, E)$  with  $|U| = |V| = n$ , the rank of the first homotopy group  $\pi_1(\Gamma_G)$  of its communication matrix  $\Gamma_G$  is  $\Theta(1/CC(G))$ , where  $CC(G)$  is the deterministic communication complexity of disjointness on  $G$ .

#### Rationale

Homotopy invariants capture global structural constraints of communication matrices, which may correlate with inherent complexity barriers. The fundamental group's rank reflects non-trivial loops in the matrix's topology, potentially mirroring the information-theoretic cost of coordination.

#### Novelty

- Judge: NOVEL over 0 arXiv hits

#### Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1218782.py", line 86, in <module>
    result = run_trial(seed)
             ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1218782.py", line 68, in run_trial
    M = clique_incidence_matrix(G)
        ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1218782.py", line 37, in clique_incidence
    M[u][v+n] = 1
    ~~~~^
```

IndexError: list assignment index out of range

## Judge reasoning

Test crashed with IndexError, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Fix the clique\_incidence\_matrix implementation to handle index bounds properly

---

## Diophantine Approximation Error Inversely Proportional to Communication Discrepancy in XOR-Functions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Diophantine Approximation  $\times$  Communication Complexity of XOR-Functions
- **Recorded:** 2026-05-11 00:27 UTC
- **Entry ID:** e98d1643eeaa

## Statement

For an XOR-function  $f: \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}$ , let  $D(f)$  denote its communication discrepancy and  $\varepsilon(f)$  denote the minimal approximation error of its Fourier coefficients by rational numbers with denominator  $\leq n^2$ . Then  $D(f) \geq \Omega(1/\varepsilon(f) \cdot \log n)$ .

## Rationale

Diophantine approximation quantifies how well real numbers can be approximated by rationals, while discrepancy measures imbalance in communication matrices. Linking them could reveal structural constraints on XOR-functions' Fourier spectra, potentially separating complexity classes via approximation-theoretic barriers.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 1.91s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b764fd6.py", line 101, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b764fd6.py", line 67, in run_trial
    D_f = communication_discrepancy(xor_function, n)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b764fd6.py", line 52, in communication
    D += abs(f[(x, y)] - f[(x + 1) % (2**n)][(y + 1) % (2**n)])
           ~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

KeyError: 1

## Judge reasoning

Test crashed with KeyError during computation of communication discrepancy, preventing validation of conjecture | next: Fix the KeyError in communication\_discrepancy function by ensuring all (x,y) pairs are properly initialized in the XOR-function's domain

---

## Hypergraph Matching Complexity for Monotone DNF

- **Verdict:** INCONCLUSIVE
- **Bridge:** Hypergraph Theory × Monotone DNF Formulas
- **Recorded:** 2026-05-11 00:37 UTC
- **Entry ID:** 5de245e2396b

### Statement

For any monotone DNF formula with  $m$  terms over  $n$  variables, the maximum matching size  $\mu$  satisfies: (i)  $\mu \leq \log n + 2\sqrt{m}$  for  $m \leq n^2$ , (ii)  $\mu \geq n/2$  for  $k$ -CLIQUE DNFs with  $k \geq 2\sqrt{n}$ .

### Rationale

Hypergraph matching captures structural limitations of DNFs; its growth rate could expose inherent complexity gaps between polynomial-size DNFs and  $k$ -CLIQUE's exponential requirements. The conjecture links hypergraph expansion properties to circuit size lower bounds.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5734af53.py", line 94, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5734af53.py", line 68, in run_trial
    matching_size = len(greedy_matching(hypergraph))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5734af53.py", line 50, in greedy_matching
    matching.append(list(hypergraph_copy[max_edge_index]))
                     ~~~~~^~~~~~
```

TypeError: list indices must be integers or slices, not NoneType

### Judge reasoning

Test crashed due to TypeError, preventing result collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the hypergraph indexing error in test\_5734af53.py and re-run experiments

---

## Additive Energy Threshold for ACC<sup>0</sup> Circuit Size in CNF Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Additive Combinatorics × ACC<sup>0</sup> Circuit Size for CNF Formulas
- **Recorded:** 2026-05-11 00:43 UTC
- **Entry ID:** 0553f6013ae8

## Statement

For any CNF formula  $\Phi$  with  $n$  variables, if the additive energy of the set of satisfying assignments  $S$  is  $\Omega(n^3)$ , then  $\Phi$  cannot be computed by an  $\text{ACC}^0$  circuit of size  $o(n^2)$ .

## Rationale

Additive energy measures the structure of the solution set;  $\text{ACC}^0$  circuits have limited capacity to capture such complex structures, leading to a lower bound on circuit size.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89dca52f.py", line 83, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89dca52f.py", line 63, in run_trial
    energy = add_energy(cnf, assignment)
             ^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89dca52f.py", line 49, in add_energy
    for x, y, z, w in combinations(range(n), 4):
                                   ^
```

NameError: name 'n' is not defined

## Judge reasoning

Test crashed due to undefined variable 'n' in the add\_energy function, preventing data collection. |  
next: Fix the NameError in test\_89dca52f.py by defining 'n' before using it in combinations

---

## SOS Refutation Size Bounded by Curve Genus of CSP Constraints

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Geometry (Curves over Finite Fields) × SOS Refutation Size for CSP Instances
- **Recorded:** 2026-05-11 00:50 UTC
- **Entry ID:** 66fe027cc0e4

## Statement

For a CSP instance over  $\text{GF}(2^n)$  with constraints defining a smooth projective curve  $C$  of genus  $g$ , the SOS refutation size is at most  $2^{O(g \log n)}$ . Conversely, if the curve has genus  $g \geq \log n$ , the SOS refutation size exceeds  $2^{\Omega(g)}$ .

## Rationale

The genus captures the topological complexity of the solution space, influencing the number of moments required for SOS to capture contradictions. High-genus curves may encode inherently complex interactions between constraints, necessitating exponential refutation depth.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 0, elapsed: 0.03s

```
9465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
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TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
RESULT: SUPPORTED mean=12.896247699465729 std=0.0 support_fraction=1.0
```

## Judge reasoning

Safety rail: critic\_challenge | original judge: All 30 tested instances satisfy the conjecture with consistent metric values and no counterexamples found. | next: Test larger genus values and higher  $n$  to validate the exponential bounds

# Tutte Polynomial Evaluation Bounded by Monotone k-CLIQUE Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tutte Polynomial Theory  $\times$  Monotone Circuit Size for k-CLIQUE
- **Recorded:** 2026-05-11 01:05 UTC
- **Entry ID:** 954a5c8e8ffd

### Statement

For any  $k$ -CLIQUE instance on  $n$  vertices, the Tutte polynomial evaluated at  $(2, 1)$  satisfies  $T(2,1) \geq \Omega(n^{\lfloor k/2 \rfloor})$  and  $T(1,2) \geq \Omega(n^{\lfloor k/2 \rfloor})$ , where  $T(x,y)$  is the Tutte polynomial of the graphic matroid induced by the complete graph  $K_n$ .

## Rationale

The Tutte polynomial encodes matroid structure, and its evaluation at  $(2,1)$  and  $(1,2)$  captures connectivity properties. For  $k$ -CLIQUE, these values may correlate with the exponential growth of monotone circuit size due to the combinatorial explosion in clique enumeration.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 99, in <module>
    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 72, in run_trial
    T21, T12 = get_tutte_values(graph, n // 2 + 1)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 61, in get_tutte_val
    T21 = tutte_polynomial(graph)
    ^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 29, in tutte_polynom
    if (graph,) in memo:
    ^^^^^^^^^^^^^^^^^
```

TypeError: unhashable type: 'collections.defaultdict'

## Judge reasoning

Test crashed due to unhashable defaultdict graph input | next: Fix graph representation to use hashable structures like frozenset for memoization

---

## Young Tableau Count Exponential Gap in Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Combinatorics (Standard Young Tableau Count)  $\times$  Communication Complexity (Disjointness Problem)
- **Recorded:** 2026-05-11 01:20 UTC
- **Entry ID:** 2b7d955e959e

## Statement

For a random disjointness instance with  $n$  variables, the number of standard Young tableaux of shape  $\lambda$  (where  $\lambda$  is the partition derived from the communication matrix) is exponentially larger than for the corresponding determinant matrix, implying a lower bound on the communication complexity.

## Rationale

The exponential gap in tableau counts reflects the combinatorial structure of the communication matrix, which could be leveraged to derive lower bounds via representation-theoretic methods.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 86, in <module>
```

```

    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 66, in run_trial
    det_tableau_count = count_standard_young_tableaux(det_partition)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 58, in count_standard_young_tableaux
    return hook_length_formula(partition)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 26, in hook_length_formula
    for cell, size in enumerate(row):
    ^^^^^^^^^^^^^^^^^
TypeError: 'int' object is not iterable

```

## Judge reasoning

Test crashed due to `TypeError`, preventing data collection. The crash indicates a code bug rather than a conjecture failure. | next: Fix the `hook_length_formula` implementation to handle partitions correctly

---

## Fourier Coefficient Sum Inverse Proportional to Disjointness Discrepancy

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier Analysis of Boolean Functions × Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-11 01:40 UTC
- **Entry ID:** 982d20a21495

### Statement

For every  $n$ -variable DISJOINTNESS instance, the sum of absolute Fourier coefficients of its characteristic function satisfies  $\sum |\hat{f}(S)| = \Theta(1/\Delta)$ , where  $\Delta$  is the discrepancy of the communication matrix. Equality holds when the instance is a balanced complete bipartite graph.

### Rationale

Fourier coefficients capture correlation with parity functions, while discrepancy measures communication complexity. Their inverse proportionality suggests hidden structural alignment between spectral properties and combinatorial communication lower bounds.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.06s

```

ed": 463, "metric_name": "Fourier Coefficient Sum", "metric_value": 466.0, "instances_tested": 40, "config": {"seed": 503, "metric_name": "Fourier Coefficient Sum", "metric_value": 548.0, "instances_tested": 40},
TRIAL: {"seed": 503, "metric_name": "Fourier Coefficient Sum", "metric_value": 548.0, "instances_tested": 40},
TRIAL: {"seed": 547, "metric_name": "Fourier Coefficient Sum", "metric_value": 438.0, "instances_tested": 40},
TRIAL: {"seed": 593, "metric_name": "Fourier Coefficient Sum", "metric_value": 476.0, "instances_tested": 40},
TRIAL: {"seed": 631, "metric_name": "Fourier Coefficient Sum", "metric_value": 454.0, "instances_tested": 40},
TRIAL: {"seed": 677, "metric_name": "Fourier Coefficient Sum", "metric_value": 504.0, "instances_tested": 40},
TRIAL: {"seed": 727, "metric_name": "Fourier Coefficient Sum", "metric_value": 564.0, "instances_tested": 40},
TRIAL: {"seed": 773, "metric_name": "Fourier Coefficient Sum", "metric_value": 440.0, "instances_tested": 40},

```

```

TRIAL: {"seed": 821, "metric_name": "Fourier Coefficient Sum", "metric_value": 632.0, "instances_tested": 1, "conjecture_holds": True}
TRIAL: {"seed": 877, "metric_name": "Fourier Coefficient Sum", "metric_value": 504.0, "instances_tested": 1, "conjecture_holds": True}
TRIAL: {"seed": 929, "metric_name": "Fourier Coefficient Sum", "metric_value": 616.0, "instances_tested": 1, "conjecture_holds": True}
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=11

```

## Judge reasoning

Safety rail: critic\_challenge\_falsified | original: Test results show conjecture\_holds is False for multiple seeds with counterexample 'mapping\_undefined'. The presence of any counterexample meets the f | next: Investigate the 'mapping\_undefined' counterexample to determine specific DISJOINT-NESS instances violating the conjecture

## SOS Refutation Degree Bounded by Clause Density for Random 3-SAT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry  $\times$  3-SAT Instances
- **Recorded:** 2026-05-11 01:51 UTC
- **Entry ID:** 52c162cccd9b

## Statement

For a random 3-SAT instance with  $n$  variables and  $m$  clauses, the minimal SOS degree required to refute it is  $\Theta(\sqrt{m})$ .

## Rationale

The SOS hierarchy's refutation degree captures the algebraic complexity of constraint systems. By linking it to clause density, we expose geometric structure in SAT solving that could yield new lower bounds for refutation algorithms.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.10s

```

569495584, got inf"}
TRIAL: {"seed": 593, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 631, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 677, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 727, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 773, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 821, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 877, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}
TRIAL: {"seed": 929, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjecture_holds": False}

```

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e7c05f5.py", line 93, in <module>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])
                        ~~~~~^~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e7c05f5.py", line 93, in <genexpr>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])

```

## Judge reasoning

parse\_fail | next: NONE

---

## Schatten p-Norm Lower Bound on Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative  $L^p$  Geometry  $\times$  Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-11 01:59 UTC
- **Entry ID:** aa045d952985

### Statement

For any  $n \geq 1$ , the Schatten p-norm of the communication matrix for the DISJOINTNESS problem on n-bit inputs is  $\Omega(n)$ , where p is the dual exponent satisfying  $1/p + 1/q = 1$  for the optimal lower bound.

### Rationale

The Schatten p-norm captures the operator structure of the communication matrix, which is tightly linked to the required information-theoretic resources in multi-party communication. For DISJOINTNESS, the norm's growth with n reflects the inherent difficulty of coordinating disjointness checks.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.14s

```
nstances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Schatten p-
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 10.011753727979853, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.731746236597688, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 10.011658675891052, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.672856354946585, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 8.734210406193428, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.586306087041859, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.752608783128863, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.97251009486534, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.914781880120312, 'instances_tested': 30,
norm does not scale linearly with n'}
```

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp\_sandbox/test\_be69cc65.p

## Judge reasoning

Test crashed with ZeroDivisionError, preventing reliable results. Pre-registered support condition cannot be evaluated without valid data. | next: Fix the test to handle empty results and re-run with sufficient sample size

---

## Matroid Rank Lower Bound for Monotone k-CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory  $\times$  Monotone Circuit Size for k-CLIQUE
- **Recorded:** 2026-05-11 03:08 UTC
- **Entry ID:** cf5188cf0224

### Statement

For a monotone DNF formula  $\varphi$ , let  $M_\varphi$  be the matroid whose ground set is clauses of  $\varphi$  and rank function  $r(S) = \max\{|S'| : S' \subseteq S, S' \text{ is a collection of pairwise disjoint clauses}\}$ . Then  $r(M_\varphi) = O(\log n)$  for all  $\varphi$  with circuit size  $\leq n^c$ , but  $r(M_\varphi) = \Omega(n)$  for all k-CLIQUE instances with  $k \geq 3$ .

### Rationale

Matroid rank captures combinatorial independence, which may mirror the structural constraints of monotone circuits. High rank for k-CLIQUE suggests inherent complexity, while low rank for general formulas hints at decomposability, offering a novel angle to circuit lower bounds.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 124, elapsed: 240.02s

TIMEOUT after 240s

## Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation limits

---

## Symmetric Power Decomposition Complexity Gap for 3-SAT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Schur-Weyl Duality  $\times$  3-SAT Instances
- **Recorded:** 2026-05-11 03:39 UTC
- **Entry ID:** 9f467f497a7b

### Statement

For a 3-SAT instance with  $n$  variables, the number of irreducible components in the symmetric power decomposition of its clause-variable incidence matrix is  $\Omega(2^n)$  larger than for the corresponding determinant's decomposition.

## Rationale

Schur-Weyl duality decomposes tensor powers into irreducible representations; permanent's symmetric nature yields exponentially more components than determinant's alternating structure, exposing algebraic barriers to polynomial identity testing.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.39s

79426043297591592576425399311708450748681513839438964967856509413642281551708967083891786846523124  
TRIAL: {'metric\_name': 'Symmetric Power Decomposition Complexity Gap', 'metric\_value': 0, 'instances\_

## Judge reasoning

Test crashed before producing valid data, rendering the metric value unreliable. | next: Re-run test with increased time limits and verify code stability

---

## Specht Module Decomposition Exponential Gap Between Permanent and Determinant Polynomials

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic combinatorics (Specht modules, symmetric functions) × SOS refutation degree for polynomial CSP instances
- **Recorded:** 2026-05-11 04:26 UTC
- **Entry ID:** 5afbe3121f1c

## Statement

For random  $n$ -variable polynomial CSP instances derived from the permanent and determinant, the SOS refutation degree of the permanent instance is exponentially larger than that of the determinant instance for all  $n \geq 2$ .

## Rationale

The exponential growth in the number of Specht modules in the permanent's decomposition suggests higher structural complexity, which could correlate with increased SOS refutation requirements.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 124, elapsed: 240.05s

TIMEOUT after 240s

## Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation to complete execution

---

## Persistent Homology Gap in Read-Twice BPs for IP\_2

- **Verdict:** INCONCLUSIVE
- **Bridge:** Persistent Homology (Topological Data Analysis) × Read-Twice Branching Programs
- **Recorded:** 2026-05-11 04:32 UTC
- **Entry ID:** f6f1e06ee5ff

### Statement

For read-twice BP  $P$  computing  $IP_2$ , the maximum persistence of 1-dimensional homology classes in the simplicial complex built from  $P$ 's state transition graph satisfies  $\text{persistence}(P) \geq \Omega(n)$ . For general read-twice BPs computing any function  $f$ ,  $\text{persistence}(P) \leq O(\log \text{size}(P))$

### Rationale

Persistent homology captures topological robustness of state transition structures.  $IP_2$ 's inherent symmetry creates persistent holes in its BP structure, while general functions' transitions collapse faster. This links algebraic topology to BP size complexity.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.04s

```
dom': False, 'counterexample': 'IP_2 BP failed: persistence(P) < Ω(n)\nRandom function BP failed: pers
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 3, 'metric_value_random': 3, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 3, 'metric_value_random': 3, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 4, 'metric_value_random': 4, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 4, 'metric_value_random': 4, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 9, 'metric_value_random': 9, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 9, 'metric_value_random': 9, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 19, 'metric_value_random': 19, 'instances_
```

## Judge reasoning

Insufficient instances\_tested=2 to validate asymptotic bounds. Counterexamples may stem from small- $n$  behavior rather than general complexity. | next: Test with larger  $n$  (e.g.,  $n \geq 1000$ ) to observe asymptotic scaling of persistence metrics

---

## Free Entropy Gap in Read-Twice BP Transition Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability Theory × Read-Twice Branching Programs
- **Recorded:** 2026-05-11 04:50 UTC
- **Entry ID:** 9858410e3069

## Statement

Let  $P$  be a read-twice BP computing a function  $f: \{0,1\}^n \rightarrow \{0,1\}$ . Define  $M_P$  as the  $n \times n$  matrix where  $M_P[i][j] = (\text{number of paths from root to leaf } i \text{ that read bit } j \text{ twice})$ . The free entropy  $\phi(M_P)$  satisfies  $\phi(M_P) \leq \log(\text{size}(P)) + O(\sqrt{n})$ , but for  $IP_2$  (inner product mod 2),  $\phi(M_P) \geq \Omega(n \log n)$ .

## Rationale

Free entropy quantifies non-commutative randomness in matrix laws; read-twice BPs exhibit structured non-commutativity via repeated bit access. This gap could reveal inherent complexity in repeated access patterns, analogous to how Shannon entropy bounds communication complexity.

## Novelty

- Judge: INCONCLUSIVE over 5 arXiv hits

Top hits consulted: - [1201.0716v2] Concavification of free entropy - [1510.05486v2] Two types of branching programs with bounded repetition that cannot efficiently compute monotone 3-CNFs - [1902.03873v2] Analogues of Entropy in Bi-Free Probability Theory: Non-Microstate - [1809.06500v3] Range entropy: A bridge between signal complexity and self-similarity - [1902.03874v3] Analogues of Entropy in Bi-Free Probability Theory: Microstates

## Empirical Test

- exit code: 1, elapsed: 1.25s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5342c042.py", line 90, in <module>
    result = run_trial(seed)
             ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5342c042.py", line 64, in run_trial
    R = r_transform(M_P)
        ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5342c042.py", line 44, in r_transform
    if M[i][j] > 0:
        ~~~~^~
```

IndexError: list index out of range

## Judge reasoning

Test crashed with IndexError before producing data; cannot assess support fraction or counterexamples | next: Debug `r_transform` implementation to handle matrix indexing properly

---

## Fundamental Group Rank Inverse Proportional to Resolution Proof Size for Tseitin Formulas

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Topology (Fundamental Groups of Graphs)  $\times$  Resolution Proof Size for Tseitin Formulas
- **Recorded:** 2026-05-11 05:05 UTC
- **Entry ID:** 2905aa033f8a

## Statement

For a Tseitin formula derived from a connected expander graph with  $m$  edges and  $n$  vertices, the resolution proof size is at least  $2^{\{c * (m - n + 1)\}}$  for some absolute constant  $c > 0$ . The exponent equals the rank of the fundamental group  $\pi_1$  of the graph.

## Rationale

Expander graphs' high connectivity implies nontrivial  $\pi_1$  structure, which may encode combinatorial obstructions to resolution proofs. The exponential dependency suggests that topological complexity directly limits proof efficiency, offering a novel algebraic-topological barrier to resolution.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.10s

```
TRIAL: {"seed": 593, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 1}
TRIAL: {"seed": 631, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 1}
TRIAL: {"seed": 677, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 1}
TRIAL: {"seed": 727, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 1}
TRIAL: {"seed": 773, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 1}
TRIAL: {"seed": 821, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 1}
TRIAL: {"seed": 877, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 1}
TRIAL: {"seed": 929, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 1}
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_32d3e164.py", line 134, in <module>
    first_failing_seed = next((r['seed'] for r in results if not r['conjecture_holds']), None)
                        ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_32d3e164.py", line 134, in <genexpr>
    first_failing_seed = next((r['seed'] for r in results if not r['conjecture_holds']), None)
                        ~~~~~^
```

KeyError: 'seed'

## Judge reasoning

Test crashed before completing required trials; insufficient data to evaluate support fraction or counterexamples | next: Fix the test code's KeyError handling and re-run with complete seed coverage

---

## Real Rank of Moment Matrix for Random Max-CUT Instances is Linear in $n$

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry (Moment Matrices, Positivstellensatz) × SOS Degree for Max-CUT
- **Recorded:** 2026-05-11 05:26 UTC
- **Entry ID:** d71faf6d27d8

## Statement

For a random Max-CUT instance with  $n$  variables and  $m$  clauses, the real rank of its moment matrix is  $\Theta(n)$ , implying that the minimal SOS degree required to approximate the maximum cut is  $\Theta(n)$ .

## Rationale

The real rank of the moment matrix captures the complexity of the polynomial system defining the Max-CUT instance. A linear real rank suggests that the SOS hierarchy requires linear degree to capture the solution, which could relate to the inherent structure of the problem.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96blab55.py", line 109, in <module>
    result = run_trial(seed)
            ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96blab55.py", line 82, in run_trial
    rank = matrix_rank(M)
          ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96blab55.py", line 37, in matrix_rank
    A = gaussian_elimination(A)
      ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96blab55.py", line 30, in gaussian_elim
    factor = A[j][i] / A[i][i]
            ~~~~~^~~~~~
```

ZeroDivisionError: float division by zero

## Judge reasoning

Test crashed with ZeroDivisionError, preventing evaluation of rank properties | next: Fix numerical stability in Gaussian elimination or test with different parameter ranges

---

## Finite-Field Rank Threshold for ACC<sup>0</sup> Circuit Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Linear Algebra over Finite Fields × ACC<sup>0</sup> Circuit Size
- **Recorded:** 2026-05-11 05:56 UTC
- **Entry ID:** de7db3666529

## Statement

Let  $f: \{0,1\}^n \rightarrow \{0,1\}$  be a Boolean function represented by a  $\text{GF}(2)$ -matrix  $M$  of rank  $r$ . Then, if  $r > n^c$  for some  $c \in (0,1)$ ,  $f$  requires ACC<sup>0</sup> circuits of size  $\Omega(n^{\frac{1}{c}})$

## Rationale

Finite-field rank captures algebraic structure that may inherently resist ACC<sup>0</sup> computation. The conjecture links matrix rank to circuit size via a power-law threshold, offering a new algebraic barrier for explicit functions.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8796f7bd.py", line 91, in <module>
    result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8796f7bd.py", line 53, in run_trial
    r = rank(matrix)
      ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8796f7bd.py", line 39, in rank
    reduced_matrix = gaussian_elimination(matrix)
                    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8796f7bd.py", line 30, in gaussian_elim
    matrix[i][j] /= factor
```

```
File "/usr/lib/python3.12/fractions.py", line 615, in forward
    return monomorphic_operator(a, b)
           ~~~~~
```

```
File "/usr/lib/python3.12/fractions.py", line 763, in _div
    raise ZeroDivisionError('Fraction(%s, 0)' % db)
```

ZeroDivisionError: Fraction(1, 0)

## Judge reasoning

Test crashed during Gaussian elimination due to ZeroDivisionError, preventing data collection | next: Fix GF(2) arithmetic implementation to use bitwise operations instead of Fraction class

---

## Secant Rank Lower Bound for Disjointness Tensors

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Geometry (Secant Varieties of Veronese Reembeddings) × Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-11 06:02 UTC
- **Entry ID:** 3b212abe5c22

## Statement

For a DISJOINTNESS matrix  $M \in \{0,1\}^{\{n \times n\}}$ , let  $T_M \in \mathbb{R}^{\{n \times n \times 2\}}$  be the tensor formed by stacking  $M$  and its transpose. The secant rank of  $T_M$  is  $\Omega(n)$ .

## Rationale

Secant rank captures the minimal number of rank-1 tensors needed to span  $T_M$ . High secant rank implies geometric complexity, which may correlate with communication complexity via the tensor product structure of DISJOINTNESS.

## Novelty

- Judge: NOVEL over 3 arXiv hits

Top hits consulted: - [1011.5867v2] Secant Varieties of Segre-Veronese Varieties - [1012.3563v4] Secant varieties to high degree Veronese reembeddings, catalecticant matrices and smoothable Gorenstein schemes - [1503.01099v2] Singularities of secant varieties

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 81, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 66, in run_trial
    secant_rank_value = secant_rank(T_M)
                       ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 53, in secant_rank
    U, V = rank_1_decomposition(T)
           ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 44, in rank_1_decomp
    U_new = matrix_multiply(U, T)
            ^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 37, in matrix_multiply
    C[i][j] += A[i][l] * B[l][j]
               ^~~~~~^~~~~~
```

TypeError: can't multiply sequence by non-int of type 'float'

## Judge reasoning

Test crashed due to a type error in matrix multiplication, preventing any data collection. The pre-registered support condition cannot be evaluated without successful trials. | next: Fix the matrix\_multiply function's type handling and re-run tests with multiple seeds to collect empirical evidence.

---

## Real Dimension of Parity-Constraint Variety Bounds $AC^0$ Depth

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry  $\times$   $AC^0$  Circuits Computing PARITY
- **Recorded:** 2026-05-11 06:31 UTC
- **Entry ID:** 5db94ffcbe67

## Statement

For any  $AC^0$  circuit  $C$  computing PARITY on  $n$  inputs, the real dimension of the semialgebraic set defined by  $C$ 's constraints is  $\Omega(\log \text{size}(C))$ .

## Rationale

PARITY's linear structure forces its constraint variety to have low real dimension, but  $AC^0$  circuits with deeper structure may encode higher-dimensional varieties. This conjecture links algebraic geometry's real dimension to circuit complexity via a concrete mapping of gates to polynomial inequalities.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.27s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_833e0728.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_833e0728.py", line 61, in run_trial
    circuit = generate_ac0_circuit(n)
             ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_833e0728.py", line 27, in generate_ac0_
    inputs = [random.randint(0, i-1) for _ in range(random.randint(2, 3))]
             ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(0, 0)
```

## Judge reasoning

Test crashed due to invalid randrange parameters, preventing data collection | next: Fix the generate\_ac0\_circuit function to ensure valid random range parameters

---

## Cheeger Constant Inverse Proportional to Extended Frege Proof Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Spectral Geometry  $\times$  Extended Frege Proof Size
- **Recorded:** 2026-05-11 07:16 UTC
- **Entry ID:** bd4677253c0f

## Statement

For a 3-CNF instance  $\Phi$  with  $n$  variables, the Cheeger constant  $h(\Phi)$  of its hypergraph representation satisfies  $h(\Phi) = \Theta(1 / \text{size}(\Pi))$ , where  $\Pi$  is the minimal Extended Frege proof of  $\Phi$ .

## Rationale

The Cheeger constant, a spectral geometry invariant, captures the expansion properties of the hypergraph. In proof complexity, expansion properties might correlate with the required proof size, as highly connected structures may necessitate longer proofs.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5952b1b6.py", line 103, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5952b1b6.py", line 86, in run_trial
    h_phi = cheeger_constant(adj_list)
            ^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5952b1b6.py", line 52, in cheeger_constant
    laplacian = [[d - A[i][j] for j, d in enumerate(degree)] + [0] * (n -
len(adj_list[i])) for i, adj_list in enumerate(adj_list)]
                ^
```

NameError: name 'A' is not defined

## Judge reasoning

Test crashed due to undefined variable 'A' in cheeger\_constant calculation, preventing data collection | next: Fix the NameError in the cheeger\_constant function by properly defining adjacency matrix 'A' before computing Laplacian

---

## Communication Matrix Rank Lower Bounds ABP Size for Boolean Functions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matrix Rank over Finite Fields  $\times$  ABP Size
- **Recorded:** 2026-05-11 07:42 UTC
- **Entry ID:** 0071c0003c24

## Statement

For any Boolean function  $f: \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}$ , the size of the minimal ABP computing  $f$  is at least the rank of its communication matrix over  $\text{GF}(2)$ .

## Rationale

The rank of the communication matrix captures linear dependencies in the function's behavior, which may impose structural constraints on the ABP's required size. ABPs inherently model computations as polynomial sequences, and low-rank matrices might correspond to more structured polynomials, potentially requiring larger ABPs to capture the necessary complexity.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 124, elapsed: 240.33s

TIMEOUT after 240s

## Judge reasoning

Test timed out without producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and debug logging to identify crash cause

---

## Invariant Generator Count Bounds SOS Refutation Degree for Symmetric CSPs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Invariant Theory × SOS Refutation Degree for CSPs
- **Recorded:** 2026-05-11 08:00 UTC
- **Entry ID:** 6972f4aaf7c4

### Statement

For a CSP instance with symmetry group  $G$ , the SOS refutation degree is at most the number of invariant generators of its constraint polynomials under  $G$ . Formally, if  $P$  is a polynomial ideal defining the CSP's feasible solutions, then  $\deg_{\text{SOS}}(P) \leq |\text{InvGen}(P, G)|$ .

### Rationale

Symmetries in CSPs often reduce the effective complexity by collapsing orbits of variables. Invariant theory provides tools to decompose constraint polynomials into symmetry-invariant components, which may directly limit the SOS hierarchy's required degree via orbit-stabilizer principles.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_lab6691.py", line 117, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_lab6691.py", line 94, in run_trial
    if is_invariant(poly, G):
       ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_lab6691.py", line 60, in is_invariant
    if poly[i] != poly[g[0][i]]:
       ~~~~~^
```

TypeError: 'int' object is not subscriptable

## Judge reasoning

Test crashed due to TypeError in subscripting integer object, preventing data collection | next: Fix the is\_invariant function to handle polynomial representations correctly

---

# Newton Polytope Volume Inverse Proportional to SOS Degree for Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Geometry (Newton Polytopes) × SOS Degree for Max-CUT Instances
- **Recorded:** 2026-05-11 08:32 UTC
- **Entry ID:** 9dba9c55f385

### Statement

For a Max-CUT instance with  $n$  variables, the volume of the Newton polytope of its corresponding degree- $d$  SOS relaxation polynomial is  $\Theta(1/d^2)$ . The volume is inversely proportional to  $d^2$ , with the constant depending on the instance's edge density.

## Rationale

Newton polytopes encode the monomial structure of polynomials, which directly reflects the constraint geometry of SOS relaxations. The volume captures the ‘spread’ of monomials, which may limit the efficiency of higher-degree relaxations by increasing the number of terms in the Gram matrix.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 0, elapsed: 0.04s

```
Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
RESULT: SUPPORTED mean=0.0 std=0.0 support fraction=1.0
```

## Judge reasoning

Tested instances show perfect support but suffer from  $n \leq 15$  (trivial scaling), metric saturation at 0, and potential selection bias. Pre-registered criteria require  $\geq 80\%$  seeds, which is met, but critic challenges invalidate confidence. | next: Test with larger  $n$  ( $\geq 30$ ) and adversarial edge density configurations to probe non-trivial volume scaling behavior

## Krull Dimension Lower Bounds SOS Refutation Degree for 3-SAT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Geometry (Krull Dimension of Ideals) × SOS Refutation Degree for 3-SAT Instances
- **Recorded:** 2026-05-11 08:45 UTC
- **Entry ID:** e6b8ed98a856

### Statement

For a 3-SAT instance with  $m$  clauses over  $n$  variables, the SOS refutation degree required to prove unsatisfiability is at least the Krull dimension of the ideal generated by the clause polynomials (each clause is mapped to a polynomial via  $(1 - x_i)(1 - x_j)(1 - x_k) = 0$  for a clause  $(x_i \vee x_j \vee x_k)$ ).

### Rationale

The Krull dimension captures the ‘geometric complexity’ of the solution space’s algebraic variety. SOS refutations must account for this structure, suggesting a lower bound tied to the variety’s intrinsic dimension. This bridges algebraic geometry with proof complexity via polynomial optimization.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_097a0593.py", line 124, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_097a0593.py", line 103, in run_trial
    m = random.randint(n, min(2 * n, 30))
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(33, 31)
```

### Judge reasoning

Test crashed due to invalid randrange parameters (33, 31), preventing data collection. Support condition cannot be evaluated without valid results. | next: Fix the test parameters to ensure valid randrange ranges (stop >= start) and re-run with proper seed handling

---

## Finite-Field Rank Threshold for ACC<sup>0</sup> Circuit Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Finite Field Geometry × ACC<sup>0</sup> Circuit Size for Sipser Function
- **Recorded:** 2026-05-11 10:35 UTC
- **Entry ID:** ac07c338962c

**Statement**

For the Sipser function on  $n$  variables, the rank of its truth-table matrix over  $\text{GF}(2)$  is  $\Omega(2^{\{n/2\}})$ , and this rank is exponentially larger than the minimal  $\text{ACC}^0$  circuit size computing it.

**Rationale**

The high rank of the truth-table matrix over  $\text{GF}(2)$  reflects the function's algebraic structure, which may inherently resist efficient  $\text{ACC}^0$  computation due to the limited depth of such circuits.

**Novelty**

- Judge: NOVEL over 0 arXiv hits

**Empirical Test**

- exit code: 124, elapsed: 240.13s

TIMEOUT after 240s

**Judge reasoning**

Test timed out without producing results; insufficient data to evaluate support fraction or detect counterexamples. | next: Increase timeout duration and re-run experiment with deterministic seed configuration